

Practical Activity: Prompt Analysis and Improvement

Objective

Analyze the following prompts, identify their strengths and weaknesses, then rewrite them to produce better AI responses.

Example 1

Original Prompt

Write something about artificial intelligence.

Strengths

- Simple and easy to understand.
- Clearly identifies the topic.

Weaknesses

- Too broad.
- No target audience.
- No output format specified.
- No length requirement.
- No objective provided.

Improved Prompt

Act as an AI educator. Write a 500-word article explaining Artificial Intelligence to undergraduate computer science students. Include a definition, key applications, benefits, challenges, and future trends. Use simple language and provide real-world examples. Format the response with headings and bullet points.

Example 2

Original Prompt

Make a website for my company.

Strengths

- Specifies the desired outcome.

Weaknesses

- No company information.
- No design requirements.
- No technology stack specified.
- No functionality requirements.

Improved Prompt

Act as a senior web designer and developer. Create a responsive homepage design for AIWAZIRI LIMITED, a technology and AI consulting company. Use a modern professional style with navy blue and gold colors. Include a hero section, services section, testimonials, about section, contact form, and call-to-action buttons. Generate the HTML, CSS, and JavaScript code and ensure mobile responsiveness.

Example 3

Original Prompt

Help me study for exams.

Strengths

- States the intention.

Weaknesses

- Subject not specified.
- Exam level unknown.
- No learning preferences.
- No timeline provided.

Improved Prompt

Act as an experienced tutor. Help me prepare for my Artificial Intelligence I (AI381) final exam. Create a 14-day study plan covering search algorithms, knowledge representation, machine

learning fundamentals, and intelligent agents. Include daily learning objectives, revision exercises, and practice questions.

Example 4

Original Prompt

Create a business plan.

Strengths

- Clear business-related goal.

Weaknesses

- Industry unspecified.
- No business model details.
- No target market information.
- No financial requirements.

Improved Prompt

Act as a startup consultant. Create a comprehensive business plan for an AI-powered educational technology company in Nigeria. Include executive summary, market analysis, competitor analysis, business model, marketing strategy, operations plan, financial projections for three years, risk assessment, and growth strategy.

Example 5

Original Prompt

Design a logo.

Strengths

- Clear design request.

Weaknesses

- No brand information.

- No color preferences.
- No style direction.
- No target audience.

Improved Prompt

Act as a professional brand designer. Design a logo concept for AIWAZIRI LIMITED. The brand focuses on artificial intelligence, research, innovation, and digital transformation. Use premium gold and deep navy blue colors. The logo should communicate intelligence, trust, innovation, and professionalism. Provide three logo concepts with symbolism explanations and recommended typography.

Discussion Questions

1. What information was missing from each original prompt?
2. How did adding context improve the prompt?
3. Which improved prompt is most likely to generate a high-quality response? Why?
4. What additional constraints could be added to make the prompts even better?
5. How does specifying a role (e.g., “Act as a startup consultant”) influence AI output?

Key Takeaway

A strong prompt usually contains:

- ✓ Role Assignment
- ✓ Clear Objective
- ✓ Relevant Context
- ✓ Specific Requirements
- ✓ Constraints
- ✓ Desired Output Format
- ✓ Target Audience
- ✓ Success Criteria

Formula

Role + Task + Context + Requirements + Constraints + Output Format = Effective Prompt

This activity helps participants move from vague prompts to precise, high-quality instructions that consistently produce better AI-generated results.